Report R20 — The External Probe, Both Encoders, All the Slides

The external site signature is real under both encoders but only *perfect* under one. And R19's input had silently omitted 51 of the archive's 80 HTMCP slides.

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Summary

Two loose ends, both closed here.

R18 found that subtype site-leakage reproduces under an encoder trained on natural images while methylation leakage does not. That dissociation turns the external probe — EAGLE against HTMCP, which R19 measured at balanced accuracy 1.000 — from a settled result into an open question, and R18's own audit listed it as un-run.

Separately, while building the second encoder's features it became clear that **R19's probe used 78 slides where the archive holds 129**. Its input had been assembled from a partial local staging, so **51 of the 80 HTMCP slides were absent**, and neither R19 nor Paper 1 records that.

Encoder	Slide set	n	Bal. acc.	Null mean	Perm. p
`owkin/phikon-v2`	SET_R19 (49 + 29)	78	**1.0000**	0.5072	0.0050
`owkin/phikon-v2`	SET_HE (49 + 27)	76	**1.0000**	0.5023	0.0050
`owkin/phikon-v2`	SET_ALL (49 + 80)	129	**1.0000**	0.5011	0.0050
`facebook/dinov2-large`	SET_R19	78	0.8934	0.5070	0.0050
`facebook/dinov2-large`	SET_HE	76	**0.8768**	0.4990	0.0050
`facebook/dinov2-large`	SET_ALL	129	0.9444	0.5020	0.0050

Corrected 2026-08-19 (R22). Every balanced accuracy in this report is a single cross-validation split. R22 measured the same quantity over 25 splits: dinov2-large is 0.9176 ± 0.0247 (range 0.8583–0.9528), so the 0.8768 below sits 1.6 standard deviations below its own mean and the Phikon-v2-vs-dinov2 gap is **1.000 versus 0.918**, not 1.000 versus 0.877. Phikon-v2 is genuinely saturated (1.0000 ± 0.0000 over 25 splits). The verdict and the substantive conclusion stand; the magnitude was overstated. R22 also adds UNI (0.9937), H-optimus-0 (0.9961) and Virchow2 (0.9601), all above their nulls.

Verdict: EXTERNAL_SIGNATURE_WEAKER_UNDER_SECOND_ENCODER, from a rule fixed in the config and hashed at `ab01b9847dedcfcf...` before either encoder's arms were read. The primary arm was pre-declared as `dinov2-large` on the H&E-only set.

Two things follow, and they pull in different directions.

The signature is not an artefact of histology pretraining. 0.8768 against a null of 0.4990 at $p = 0.0050$ is a large, highly significant separation from a model that has never seen a pathology slide. Two independent institutions are distinguishable from average tile content alone, whatever the encoder.

But "perfectly separable" is a Phikon-v2 statement. Phikon-v2 reaches 1.000 and dinov2-large reaches 0.877 on the same 76 slides with the same probe. The *existence* of the signature is encoder-independent; its *magnitude* is not. This is the same pattern R18 found — a histology-specialised representation encodes institutional character more completely — and Paper 1's word "perfectly" now needs an encoder attached to it.

1 R19's missing 51 slides, and why the number survived anyway

The good news first: Phikon-v2 returns 1.0000 on all three sets — R19's 78, the full 76-slide H&E set, and all 129. So the undocumented subset did *not* bias R19's headline. The defect is one of provenance, not of correctness.

It is still a defect. A file assembled from whatever happened to be staged locally, describing itself as "two external cohorts", is the third problem found in that same lineage:

1. the two external probe rows had no persisted artefact at all (R19),
2. the input named `external_he_mean.npz` contained three non-H&E slides (R19), and
3. that input silently held 29 of 80 HTMCP slides (this report).

All three are the same species: an intermediate file whose name asserted a property nobody re-checked. Gate G2 here is written inversely — it asserts the omission is present and that the count is exactly 51 — so a silent repair of the input file fails the gate rather than passing quietly.

2 Why SET_ALL is not a result

SET_ALL mixes immunohistochemistry with H&E. A probe asked to tell two institutions apart when one contributes 53 IHC slides can win on stain, so its numbers are an upper bound and are reported only as one. Note that dinov2-large is *higher* on SET_ALL (0.9444) than on H&E alone (0.8768), which is exactly what a stain shortcut looks like: adding IHC makes the task easier rather than harder. R12's audit predicted this failure mode in writing; SET_HE is the measurement.

3 Limitations

Two institutions, 76 H&E slides. This bounds the existence of a cross-institution signature. It is not a powered external benchmark and is not presented as one.

Slide-level mean features. The probe operates on mean-pooled tile embeddings, so it measures what is linearly recoverable from average tile content, not what the attention-MIL sees.

Stain from filename suffixes. `HE`, `CHR`, `P16` are the only stain field available. Source mislabelling would propagate.

Two encoders is not a survey. Same limitation as R18. UNI, H-optimus-0 and Virchow2 sit behind licence acceptance that was not ours to give.

Magnitudes across encoders are not strictly comparable. dinov2-large is weaker at every histology task measured here, so its lower separation is expected in direction. What the arms establish is that the signature clears a permutation null under both, not that 1.000 and 0.877 are commensurable quantities.

Reproducing this report

`evidence/m17_external_probe_two_encoders.py` (the whole run), `evidence/config.yaml` (pre-declared rules, SHA-256 `ab01b9847dedcfcf...`), `evidence/gates.json` (5 gates), `evidence/results.json`, `evidence/log.jsonl`, `evidence/ph_ext_mean.npz` and `evidence/dv_ext_mean.npz` (the two 129×1024 mean-embedding matrices, with cohort and H&E flags).